II. THE THEORY OF SYSTEMS IN BIOLOGY AND MEDICINE

II.1. The meaning and delimitation of system notion
II.2. Systems classification
II.3. Main general features of the biological systems
II.4. Conditions and ways of approaching the systems for their study
II.5. Biological thermodynamics
II.6. Cybernetics and biocybernetics
II.7. Complex systems
II.8. Complexity and chaos in biosystems
II.9. What is life?

II.1. The meaning and delimitation of system notion

One of the used methods for study in biology and medicine is ***the systemic method.*** This method is concerned precisely with the organizational aspect of life and rests mainly upon the idea that all matter, be it nonliving or living, is organized in hierarchical systems. So far the first tendency prevails, and there is a lot of data, especially in the domain of molecular biology, which can't be used partially in the medical practice. The assays of unifying all the data are very few. The most successful trial to offer basis for global explanation of some functions from the living matter (human body) in normal and pathological conditions, is represented by the ***theory of systems***. This theory has extended, by nowadays, over all domains of since, including technique, in spite of its conceiving by a biologist.

The idea of a ***general theory of systems*** was advanced by ***Ludwig von Bertalanffy*** in 1938, in his lectures from Vienna University and the concept was used later, in 1951, in scientific publications. The theory consider that '*modern biology*

must consider the living organism as a systemic object', the system being delimitated like '*a complex of elements which are in interaction one which each others'*. The purpose of this theory is to express the general features (properties, characteristics) of the systems, no matter what is their structure or the nature of their elements.

Other definitions of the notion of system could be:

- ***Amosov, 1965***: '*A system represents a certain quantity of elements which can be either identical or different and which are connected one with the others, the connection being governed by certain laws'*.
- ***A more didactical definition***: *a system is a collection of components arranged and interconnected in a definite way. The elements can be: biological, physical, chemical or a combination of all three.*

Essential is that the system has new features which are different in comparison with the features of the isolated elements. For example, a molecule is a system made by atoms connected in a special way to form a whole, a unitary system, with a certain integrity and qualitative determination; a molecule of water (made by two hydrogen atoms and one oxygen atom) has, for fewer, different properties in relation with the components atoms properties and even the characteristics of the atoms inside the molecule change, because their interactions (determining the partial ionic character of hydrogen – oxygen bond and, finally, the dipole character of water, with large consequences in more complex biological systems, where water molecule represent only an element.

II.2. Systems classification

A classification of systems is presented bellow:

Table II.2.1. Classification of systems, from the point of view of energy and matter exchange

Type of system	***Change matter***	***Change energy***	***Example***
isolated	no	no	- Hypothetical existence
close	no	yes	- A class of students at the biophysics course
open	yes	yes	- A class of students in the break of the biophysics course - All living organisms

In fact systems can be *classified*:

- In relation with *the nature of the elements and the laws*, also in relation with the aspects which are studied, the systems can be: biological, physiological, physical or biophysical systems.
- From the point of view of *their connections with other surrounding systems* there are isolated, closed, open systems.
- According to *their way of function*, the open systems are: open systems without self-regulation and open systems with self-regulation, the last ones including the cybernetic systems and biological systems.

Biological systems, physiological, biophysical, physical systems

The human organisms, like all biological living systems, are maintained in a dynamic equilibrium, i.e. they change substance and energy with the environment medium, but they rest the same, through a lot of process, through continuos assimilation and desassimilation, all processes being, in normal situations, self regulated.

All living systems are ***open systems self regulated***.

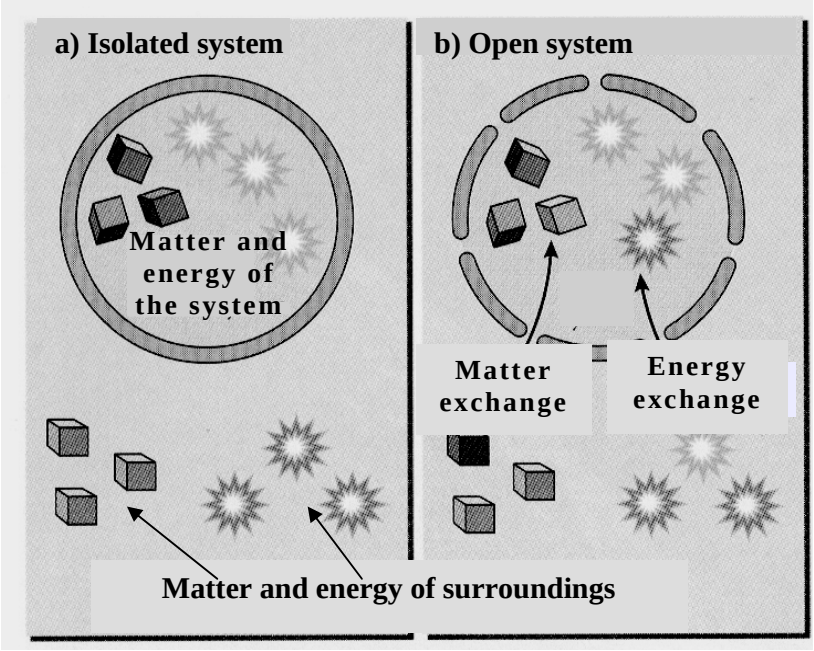


Figure II. 2.1 Isolated and open systems: a) in an isolated system matter and energy do not exchange with matter and energy outside; b) in an open system, matter and energy do exchange *(59)*

In the biological systems, the components are represented by atoms, molecules, molecular conglomerates, cells, tissues, organisms, population etc. The notion of biological systems includes also the physiological systems, which are characterized by their specific functions. Speaking about the function of a living system this system is a physiological system, e.g. if the contractile system is studied

by means of muscular contraction then a ***physiological system*** will be obtained. The same muscular system, when is studied on elastic properties criteria represents a ***biophysical system***. If we speak about measurable physical sizes of a system, this will be a ***physical system***.

The biophysical system study is on the aim of biophysics. Biophysical systems are included in the large group of biological systems. They are living structures, which are studied by means of physical features or phenomena, measurable and specific for the given system.

II. 3. Main general features of the biological systems

Biological systems have a number of features, which are specifically for them:

1. The ***informational character***: this means the ability to receive, accumulate and convert the information from the environment (primary information). The transformation of primary information in other type of signals, which can be understood by the living system, it is named decodification, the result being a secondary information (usually an action biopotential).
2. The ***integrality***: the biosystem is a whole with new structural and functional features which do not characterize each component in part; it is not only a sum of its component features. The parts functions and characteristics in the system depend on the regulation of the whole system.
3. The ***program***: this is due to the permanent change of energy and substance with the environment which determine modifications in the system but a steady state is maintained; there are three types of programs

– A program for itself (the conservation of system) which ensures the self regulation of the system.
– A lower (inferior) program of some parts of the system which can be considered, at their turn, systems; for example, the molecular program or that of the organites included in the cell system.
– A higher (superior) program, which assures the perpetuation of the species.

The lower hierarchical programs are subordinated to the laws of the immediately higher program and are coordinated by the latter; by their specific character they influence the higher program.

4. The ***dynamic equilibrium*** (steady state): there is a permanent change of substance and energy, permanent renewal, but the system rests the same, because of self-regulation mechanisms.
5. The ***self-regulation***: the byosistems have the ability to compare the effectuated answer with the given order by feed-back ways; the ways of feed-back is not specific for each parameter, being the blood, for the endocrine system, or the nerves, for the nervous system).